Date	31 October 2026
Team ID	XXXXXX
Project Name	Global Fuel Prices and Taxation Analysis (2020–2026)
Maximum Marks	4 Marks

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Functional Requirements:
Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	**Data Access & Integration**	Connect Tableau to the global fuel prices dataset (CSV / Excel extract) spanning 84 countries, 27,468 records, multi-fuel types (Petrol, Diesel, LPG), tax percentages, subsidy indicators, country income levels, and Brent crude spot prices; relate or join tables on standard geographic and temporal keys.	High
2	**Data Processing**	Clean and standardize raw records (currency conversions, missing value handling, category mapping); assign geographic roles to Country and Region fields; create calculated metrics for Net Fuel Price, Tax Amount, Subsidy Amount, and Volatility Index for analytical modeling.	High
3	**Business Analysis**	Calculate aggregate metrics and measures: Average Petrol & Diesel Retail Price, Tax Percentage Share, Net Subsidy Burden, Crude Pass-Through Correlation Coefficient, and Regional Volatility Index; aggregate calculations by Region, Fuel Type, Country Income Level, and Year (2020–2026).	High
4	**Business Insights & Analytical Outputs**	Deliver structured interactive views tailored to key stakeholder personas: • *Policy View (Dr. Sarah Mitchell):* Tax Heatmaps, Subsidy vs. Tax scatter plots, Tax Percentage vs. Price relationships. • *Supply Chain View (Marcus Thompson):* Multi-fuel price trends, Diesel distribution maps, regional volatility tracking, and predictive price forecasting. • *Research View (Prof. Elena Rodriguez):* Income-level cross-tabulations, crude correlation models, and multi-year structural comparison.	High
5	**User Interaction & Analysis**	Allow users to filter dynamically by fuel type, region, income level, and temporal range; enable cross-highlighting, dynamic sorting, interactive tooltips, and parameter-driven scenario selection across worksheets.	Medium

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
6	**Data Update & Reporting**	Support seamless data source refreshing so updated market price feeds automatically reflect across all dashboards; enable exporting views into PDF summaries, packaged workbooks (.twbx), and high-resolution images for executive briefing.	Medium

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Dashboards feature intuitive title blocks, structured legend formatting, informative hover tooltips, and global quick-filters so policy consultants, logistics managers, and researchers can analyze insights without complex technical training.
NFR-2	**Security**	Access to raw datasets and workbook configurations is restricted to authorized team members. Published visual outputs enforce view-only permissions. The dataset contains macro public commodity pricing and contains zero PII (Personally Identifiable Information).
NFR-3	**Reliability**	Calculated fields (e.g., Tax Percentage Share, Net Subsidy Amount) yield consistent mathematical outputs post-refresh. Data source validation checks confirm 100% record integrity without missing field errors.
NFR-4	**Performance**	Dashboards leverage packaged Tableau Data Extracts (.hyper) and pre-aggregated measures, guaranteeing sheet rendering and filter response times under 3 seconds across all 27,468 rows.
NFR-5	**Availability**	Published interactive workbooks remain accessible online for stakeholder reviews. Offline backup copies (.twbx format) are stored securely across local repositories for offline presentation.
NFR-6	**Scalability**	The workbook structure accommodates additional historical years (beyond 2026), new commodity fuels (e.g., CNG, Electric Tariffs), and expanded country coverage seamlessly via simple extract refreshes without requiring structural dashboard redesigns.